

SUPRIYA PAUL

Data Scientist | Machine Learning Engineer

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PROFESSIONAL SUMMARY

Analytical Data Scientist experienced in framing business problems as machine learning tasks and building predictive models on structured and unstructured datasets. Proficient in Python, SQL, Scikit-learn, TensorFlow, and XGBoost, with expertise in EDA, feature engineering, model optimization, and evaluation using metrics such as ROC-AUC. Skilled in end-to-end ML workflows, experimentation, and deploying data-driven solutions through APIs and interactive dashboards

PROFESSIONAL EXPERIENCE

- Full Stack Engineer, Omnineura AI Technologies** 06/2024 – 01/2025
Bengaluru, Karnataka
- Led frontend architecture development for **4 production web platforms**, aligning UI implementation with product goals and scalable UX standards.
 - Implemented **Redux Toolkit for state management** and integrated **AJAX-based real-time updates**, reducing bug frequency by **20%** and decreasing page reloads by **40%**.
 - Collaborated with product, design, and QA teams to deliver features, optimize frontend performance, and improve load time and **responsiveness by 25%**.
- PGT Chemistry, Dolphin HS School** 10/2021 – 03/2024
Kendrapara, Odisha
- Taught Chemistry to Class XI–XII students and provided academic guidance and mentorship.

EDUCATION

- Minor in Artificial Intelligence and Data Science, IIT Mandi** 05/2025 – 04/2026
CGPA : 7.8
- M.Sc. in Chemistry, North Orissa University** 07/2019 – 05/2021
CGPA : 8.14
- B.Sc. in Chemistry, Rama Devi University** 08/2016 – 04/2019
CGPA : 8.62

SKILLS

Programming: Python, SQL, JavaScript
Data Analysis & Preparation: EDA, data cleaning, feature engineering, missing value treatment, outlier detection, feature scaling
Machine Learning: Logistic Regression, Random Forest, Gradient Boosting (XGBoost), model evaluation (ROC-AUC, Precision, Recall, F1-score), cross-validation, hyperparameter tuning (Optuna)
Frameworks & Libraries: scikit-learn, TensorFlow, PyTorch, Pandas, NumPy, Matplotlib, Seaborn
Model Deployment: Streamlit, FastAPI, REST API integration, model inference pipelines, batch and real-time predictions
Databases: MySQL
Web Development: React.js, Tailwind CSS, Node.js, Express.js
Tools: Git, GitHub, Docker (basic), Jupyter Notebook, Jira

PROJECTS

- Healthcare Premium Prediction System** 📄, Python | Scikit-learn | XGBoost | Streamlit
Live demo 📄
- Developed **Health Insurance Premium Prediction** models using Python, Scikit-learn, and XGBoost on **2 customer segments**, applying data preprocessing and feature engineering to improve prediction accuracy.
 - Engineered a **medical risk scoring framework** by transforming multiple health conditions into numerical features, enhancing model interpretability and predictive performance.
 - Optimized XGBoost using **RandomizedSearchCV**, achieving **R² scores of 99.7% and 98.8%** with **sub-2% prediction error**, enabling highly accurate premium estimation.
- Vehicle Damage Prediction System** 📄, Python | TensorFlow/Keras | CNN | Optuna | FastAPI | Streamlit
Live demo 📄
- Developed a **vehicle damage detection system** using CNN and transfer learning models trained on **2K+ labeled images** to automate insurance inspection workflows.
 - Applied **data augmentation, regularization, and Optuna hyperparameter tuning**, achieving **~80% validation accuracy and F1-score ≈ 0.79**.
 - Deployed the model using **FastAPI with a Streamlit interface**

CERTIFICATES

- TCS NQT- IT 📄
- Web Development Certification- Edyoda 📄
- Node,Express,Mongo Certification 📄